

VALVULAR HEART DISEASE

Overview

- Valve either **stenotic** (obstructs forward flow) or **regurgitant** (backward leak).
- **Developing countries** → **rheumatic heart disease (RHD)** commonest; **industrialized** → **degenerative**. Prevalence ↑ with age (13.3% in ≥75).
- **SOB** = common symptom; acute severe SOB = valve failure.

Acute Rheumatic Fever (ARF)

- Children/young adults **5–15 yrs**. Endemic in South Asia, Africa, South America.
- **Pathogenesis**: immune-mediated delayed response to **group A streptococcus** — antigens cross-react with cardiac myosin/sarcolemma. **Aschoff nodules** = **pathognomonic** (heart only, subacute/chronic phase).
- **Clinical (2–3 wks post-strep pharyngitis)**: fever, arthritis (75%), carditis, rashes, nodules, chorea.
- **Jones criteria**: 2 major OR 1 major + 2 minor + evidence of preceding strep.
- **Carditis**: pancarditis; **soft systolic murmur (MR) very common**; **Carey Coombs murmur** (soft mid-diastolic, valvulitis); AR in 50%; tricuspid/pulmonary rarely.
- **Arthritis**: commonest major; **migratory, asymmetric, large joints** (knees/ankles/elbows/wrists), red/swollen/tender.
- **Skin**: **erythema marginatum** (<5%, trunk/proximal, never face), **subcutaneous nodules** (5–7%, extensor surfaces, >3 wks, confirm not make dx).
- **Sydenham's chorea (St Vitus dance)**: late (≥3mo), females, choreiform movements; ¼ develop chronic valve disease.
- **Investigations**: ESR/CRP (monitor), throat culture (+ve only 10–25%), **ASO titres**, echo (MR, anterior MVL prolapse).

ARF management

- **Bed rest** until temp/WBC/ESR settle.
- **Antibiotics**: single **benzathine benzylpenicillin IM** or oral penicillin ×10d to eradicate strep; penicillin-allergic → erythromycin/cephalosporin.
- **Long-term prophylaxis** (oral phenoxymethylpenicillin BD or monthly benzathine IM): until **age 21** if no carditis; **until 10 yrs after last episode or age 40** (whichever later) if residual heart disease. (Does NOT protect against IE.)
- **Aspirin** for arthritis (response in 24h confirms dx). **Glucocorticoids** for carditis/severe arthritis (faster than aspirin).

Chronic RHD

- Develops in ≥50% of those with carditis; **2/3 women**. Progressive **fibrosis**. **MV affected >90%**, then aortic, tricuspid, pulmonary. **Isolated MS ~25%; mixed MS+MR 40%**.

Mitral Stenosis (MS)

- **Almost always rheumatic**. Normal orifice ~5 cm²; symptoms when <2 cm²; severe <1 cm².
- **Pathophysiology**: ↑LA pressure → pulmonary venous congestion + SOB. Tachycardia/exercise/pregnancy poorly tolerated. **AF common** (LA dilatation) → can precipitate pulmonary oedema. PHT → RVH/RV dilatation → TR/right HF.
- **Clinical**: effort dyspnoea (dominant), fatigue, haemoptysis, **thromboembolism (esp AF)**.

- **Signs:** **loud S1 (tapping apex)**, **opening snap** (closer to S2 = more severe), **low-pitched mid-diastolic murmur** with **pre-systolic accentuation**, malar flush.
- **Management:** medical (anticoagulation, rate control digoxin/ β -blocker for AF, diuretics). **Balloon valvuloplasty = treatment of choice** if criteria met; **MVR** if substantial MR or rigid/calcified valve. (DOACs not approved; IE prophylaxis no longer routine.)

Mitral Regurgitation (MR)

- **Chronic:** gradual LA dilatation, few symptoms, slow LV dilatation (volume overload). **Acute:** rapid LA pressure rise $\rightarrow$ marked deterioration.
- **MVP (floppy valve):** common cause of mild MR; myxomatous degeneration; $\pm$ Marfan. **Mid-systolic click** $\pm$ late systolic murmur; chordal rupture $\rightarrow$ sudden severe MR; benign arrhythmias, good prognosis.
- **Management:** medical (diuretics, vasodilators, ACE/ARB; digoxin/anticoagulant if AF). **MV repair = treatment of choice for severe MR** (prevents irreversible LV damage); annuloplasty ring if CABG.

Aortic Stenosis (AS)

- Causes by age: **congenital (infancy)**, **bicuspid (young-middle age, 50–60)**, **rheumatic (30–60, + MV disease)**, **calcific/degenerative (70–90)**.
- **Pathogenesis:** $\uparrow$ pressure gradient $\rightarrow$ **LVH** $\rightarrow$ angina (even without CAD) $\rightarrow$ fixed CO $\rightarrow$ eventual LV failure.
- **3 cardinal symptoms:** **angina**, **SOB**, **syncope** (exertional). Asymptomatic for years then rapid deterioration; **untreated die within 3–5 yrs** of symptoms.
- **Signs:** **harsh ejection systolic murmur radiating to neck**, **soft S2**, **S4**; **slow-rising low-amplitude pulse (pulsus parvus et tardus)**, narrow pulse pressure, precordial thrill; Gallavardin (radiates to apex).
- **Investigations:** **echo (pivotal)** — gradient/severity; low-flow AS if poor LV; ECG LVH + strain; CT for calcification.
- **Management:** asymptomatic $\rightarrow$ monitor (every 1–2 yrs, 3–6mo if heavily calcified), treat risk factors. **Symptomatic severe AS** $\rightarrow$ **prompt AVR** (delay $\rightarrow$ sudden death). **TAVI** (transfemoral, crushes native valve) for high/inoperable risk $\rightarrow$ now extreme-to-low risk. Old age NOT a contraindication. Balloon valvuloplasty only useful in congenital.

Aortic Regurgitation (AR)

- Causes: cusp disease, infection, trauma, **aortic root dilatation**. Developing $\rightarrow$ RHD; developed $\rightarrow$ root dilation, calcific, bicuspid.
- **Pathogenesis:** LV dilates (volume overload), SV may double $\rightarrow$ pulsatile arteries $\rightarrow$ eventually LVF.
- **Clinical:** palpitation, SOB, PND, angina. **Early diastolic murmur** (left sternal edge, held expiration); systolic flow murmur ($\uparrow$ SV, not AS); **Austin Flint murmur** (functional MS). **Acute severe AR** (endocarditis cusp tear) $\rightarrow$ HF predominates, signs masked.
- **Investigations:** **Doppler echo (1st choice)**; catheterisation/aortography; MRI for aortic dilation.
- **Management:** treat cause (IE/syphilis). **AVR if symptomatic** or **EDD ≥ 55 mm**; annual echo follow-up; vasodilator (nifedipine) for HTN; aortic root replacement if Marfan/root dilatation.

Tricuspid Stenosis (TS)

- Most commonly **rheumatic** (with MS/AV disease). Systemic venous congestion: **ascites**, **peripheral oedema**, **hepatomegaly**, **prominent a wave**, **Kussmaul sign**. Low-frequency

diastolic murmur at left lower sternal border. **TTE; severe = valve area $\leq 1.0 \text{ cm}^2$** . Treat diuretics, balloon valvotomy or surgery.

Tricuspid Regurgitation (TR)

- Usually **functional** (RV dilatation from right/biventricular HF). '**Giant**' v wave in JVP, pansystolic murmur left sternal border, **pulsatile liver**, oedema, hepatomegaly. Often improves when HF treated; annuloplasty ring if undergoing MVR.

Pulmonary Stenosis (PS)

- Almost always **congenital** ($\pm$ TOF, Noonan, rubella) or carcinoid. Mild = asymptomatic; moderate $\rightarrow$ RV overload, exertional dyspnoea; severe $\rightarrow$ RV failure, cyanosis.
- **Ejection systolic murmur left upper sternum $\rightarrow$ left shoulder**, ejection click, wide split S2; severe $\rightarrow$ inaudible P2, **fixed split S2**, RV heave.
- **Doppler echo definitive**. Severe (gradient $>50 \text{ mmHg}$) $\rightarrow$ **balloon valvuloplasty** (excellent results).

Pulmonary Regurgitation (PR)

- Rare isolated; usually PA dilatation from PHT. **Graham Steell murmur** (early diastolic, hard to distinguish from AR, complicates MS). Trivial PR = normal finding.

Prosthetic valves

- **Mechanical** (ball & cage, tilting disc, bileaflet): durable but **require lifelong anticoagulation** (thromboembolism/thrombosis risk); produce clicks.
- **Biological** (pig/allograft): **no anticoagulation needed** but less durable (degenerate 7–10 yrs, worse in mitral); **best for >65 / aortic position**.
- **TAVI**: bioprosthetic via femoral, native valve compressed; avoids sternotomy, short recovery, for high-risk/elderly. **Complications: stroke 2%, heart block needing pacemaker 5–15%**.
- **Prosthetic dysfunction**: unexplained HF $\rightarrow$ urgent assessment. Mechanical $\rightarrow$ strut fracture/thrombosis. Biological $\rightarrow$ regurgitant murmur 8–10 yrs.

High-yield one-liners

- **ARF**: Jones criteria, group A strep, Aschoff nodules, migratory arthritis, Sydenham's chorea; penicillin + prophylaxis.
- **MS** = rheumatic; loud S1 + opening snap + mid-diastolic murmur + AF; balloon valvuloplasty.
- **MR severe** $\rightarrow$ MV repair (treatment of choice). **MVP** = mid-systolic click.
- **AS**: angina/SOB/syncope; ejection systolic $\rightarrow$ neck, soft S2, slow-rising pulse; symptomatic severe $\rightarrow$ AVR/TAVI; die 3–5 yrs untreated.
- **AR**: early diastolic murmur, Austin Flint, collapsing pulse; AVR if symptomatic or EDD $\geq 55 \text{ mm}$.
- **TR** = giant v wave + pulsatile liver. **PS** = congenital, balloon valvuloplasty.
- **Mechanical valve** = anticoagulate forever; bioprosthetic = no anticoagulation but less durable (>65 yrs).